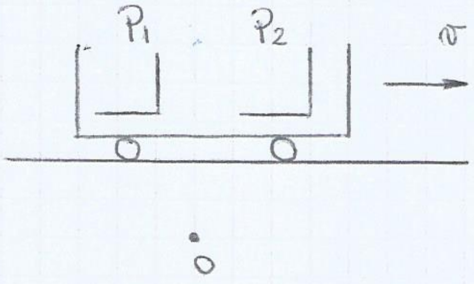
Unidad I: Cinemática.

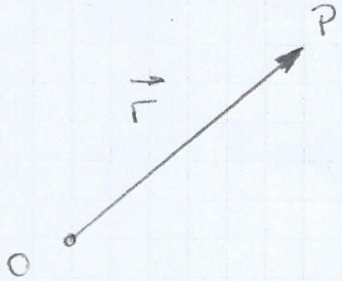
Movimiento: cambio de posición en el tiempo.

Reposo: Mantiene su posición constante



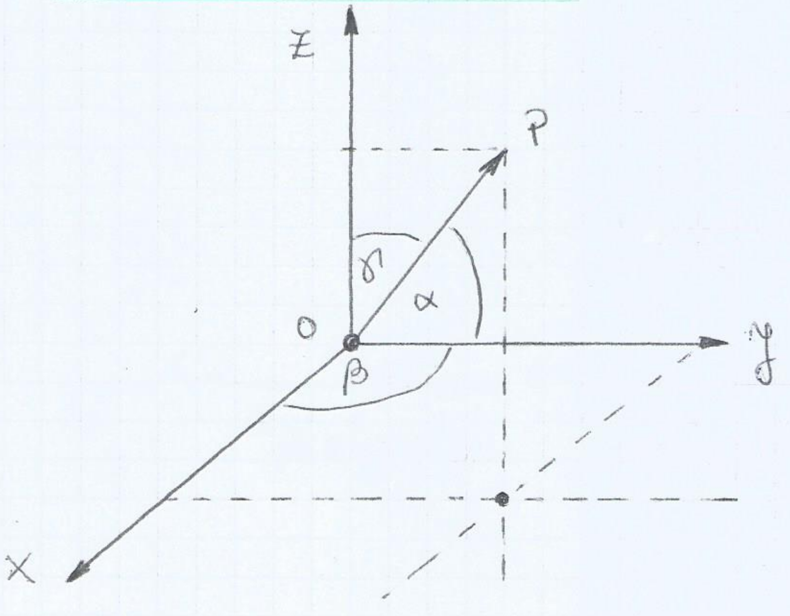
La persona en el anden observa que los pasajeros se mueven.

Definir observador: Se vincula con el sistema de referencia que se representa matematicamente con un sistema de coordenadas.



radio vector: indica la posición de P respecto de O, tambien se lo llama vector posición.

Coordenadas cartesianas:



$\vec{P} = (x_p, y_p, z_p)$

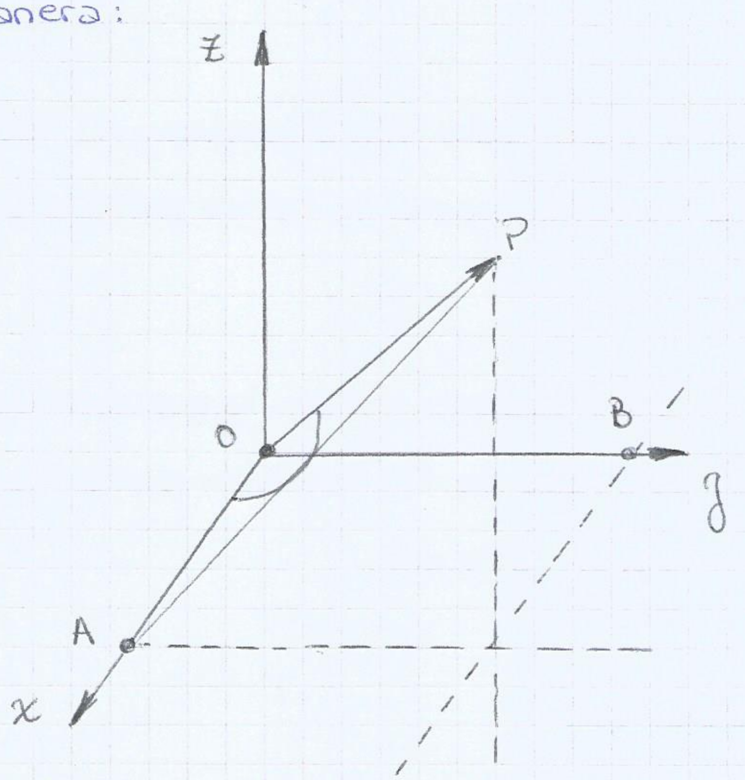
La posición es un objeto vectorial. Para definir un vector se necesita una dirección, un sentido y un módulo. Las magnitudes escalares se especifican con un solo número (temperatura, presión o energía).

Módulo de un vector: $|r|$

en $\mathbb{R}^2$ $\quad |r| = \sqrt{x_p^2 + y_p^2}$

en $\mathbb{R}^3$ $\quad |r| = \sqrt{x_p^2 + y_p^2 + z_p^2}$

Otra manera:



$\cos \alpha = \dfrac{x_p}{r}$

$\cos \beta = \dfrac{x_y}{r}$

$\cos \gamma = \dfrac{x_z}{r}$

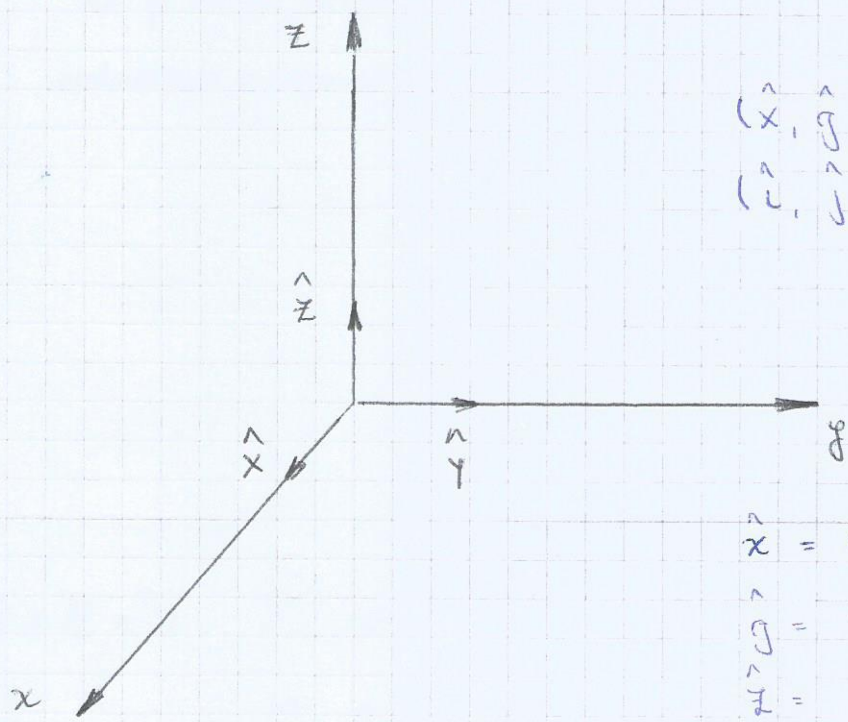
$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\frac{x_p^2}{r^2} + \frac{y_p^2}{r^2} + \frac{z_p^2}{r^2} = 1$$

$$\frac{x_p^2}{x_p^2 + y_p^2 + z_p} + \frac{y_p}{x_p^2 + y_p^2 + z_p^2} + \frac{z_p}{x_p^2 + y_p^2 + z_p^2} = 1$$

Versores fundamentales: Son vectores que tienen módulo 1

los versores fundamentales tienen dirección y sentido de los ejes coordenados.



$(\hat{x}, \hat{y}, \hat{z})$

$(\hat{i}, \hat{j}, \hat{k})$

$\hat{x} = (1, 0, 0)$

$\hat{y} = (0, 1, 0)$

$\hat{z} = (0, 0, 1)$

$\vec{A} = x_a \hat{x} + y_a \hat{y} + z_a \hat{z}$

Producto escalar: $\overline{A}$ y $\overline{B}$

$\overline{A} . \overline{B} = |A| . |B| . \cos\alpha$ $\qquad$ $\overline{A} . \overline{B} = 0 \Rightarrow \overline{A} \perp \overline{B}$

Si $\overline{A} = x_a \hat{x} + y_a \hat{y} + z_a \hat{z}$

$\overline{B} = x_b \hat{x} + y_b \hat{y} + z_b \hat{z}$

$\overline{A} . \overline{B} = x_a . x_b + y_a y_b + z_a z_b$

$\hat{x} . \hat{x} = 1$

$\hat{x} . \hat{y} = 0$

NOTA

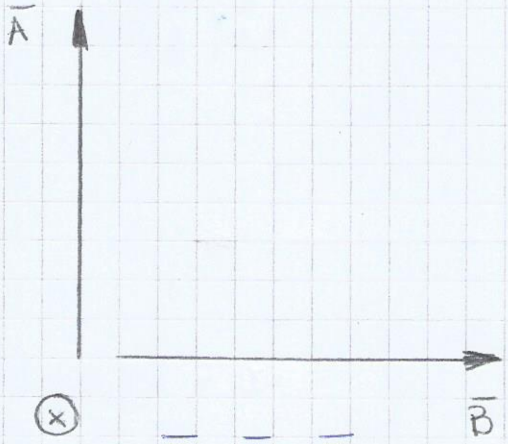
Producto vectorial : $\bar{A}, \bar{B}$

Si $\bar{C} = \bar{A} \times \bar{B}$ o $\bar{A} \wedge \bar{B}$

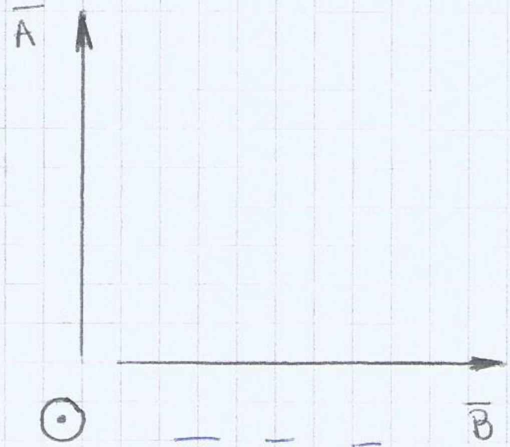
Modulo : $|A||B| \operatorname{sen}\alpha$

Dirección : $\bar{C} \perp$ plano que contiene a $\bar{A}$ y $\bar{B}$

Sentido : Siguiendo la regla de la mano derecha.



$\bar{C} = \bar{A} \times \bar{B}$



$\bar{C} = \bar{B} \times \bar{A}$

Si $\bar{A} = x_a \hat{x} + y_a \hat{y} + z_a \hat{z}$

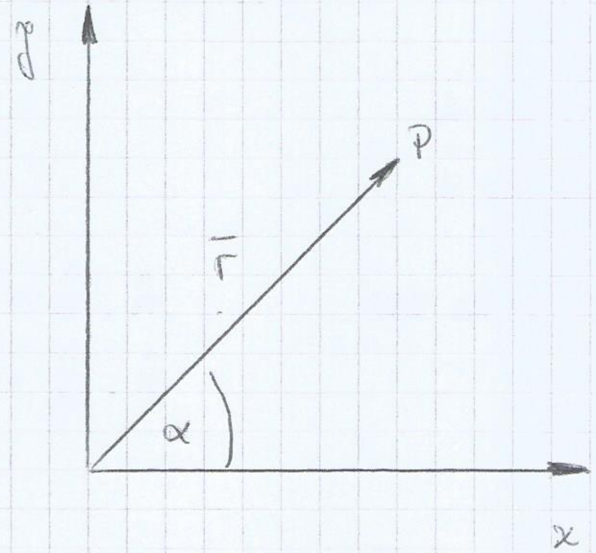
$\bar{B} = x_b \hat{x} + y_b \hat{y} + z_b \hat{z}$

$\hat{x} \times \hat{x} = 0$ $\quad$ $\hat{y} \times \hat{z} = \hat{x}$

$\hat{x} \times \hat{y} = \hat{z}$ $\quad$ $\hat{z} \times \hat{x} = \hat{y}$

$$\bar{A} \times \bar{B} = (y_a z_b - z_a y_b)\hat{x} + (z_a x_b - x_a z_b)\hat{y} + (x_a y_b - x_b y_a)\hat{z}$$

Coordenadas polares : (α, r)



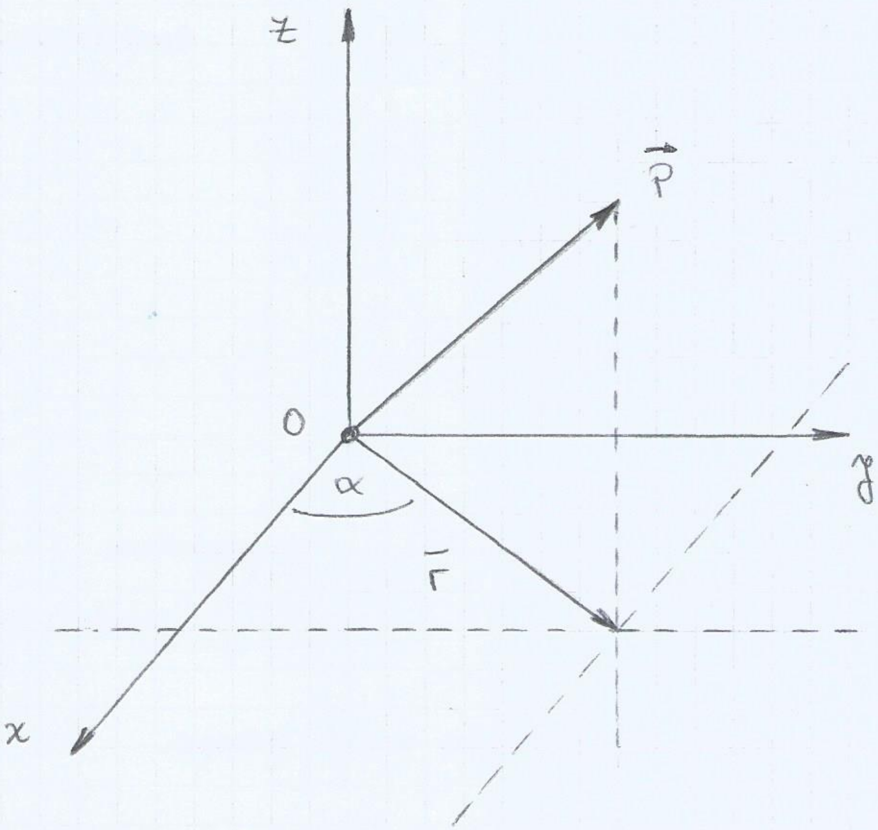
$\bar{r} = \sqrt{x_p^2 + y_p^2}$

$x_p = r \cos\alpha$

$y_p = r \operatorname{sen}\alpha$

$\operatorname{tg}\alpha = \dfrac{y_p}{x_p}$

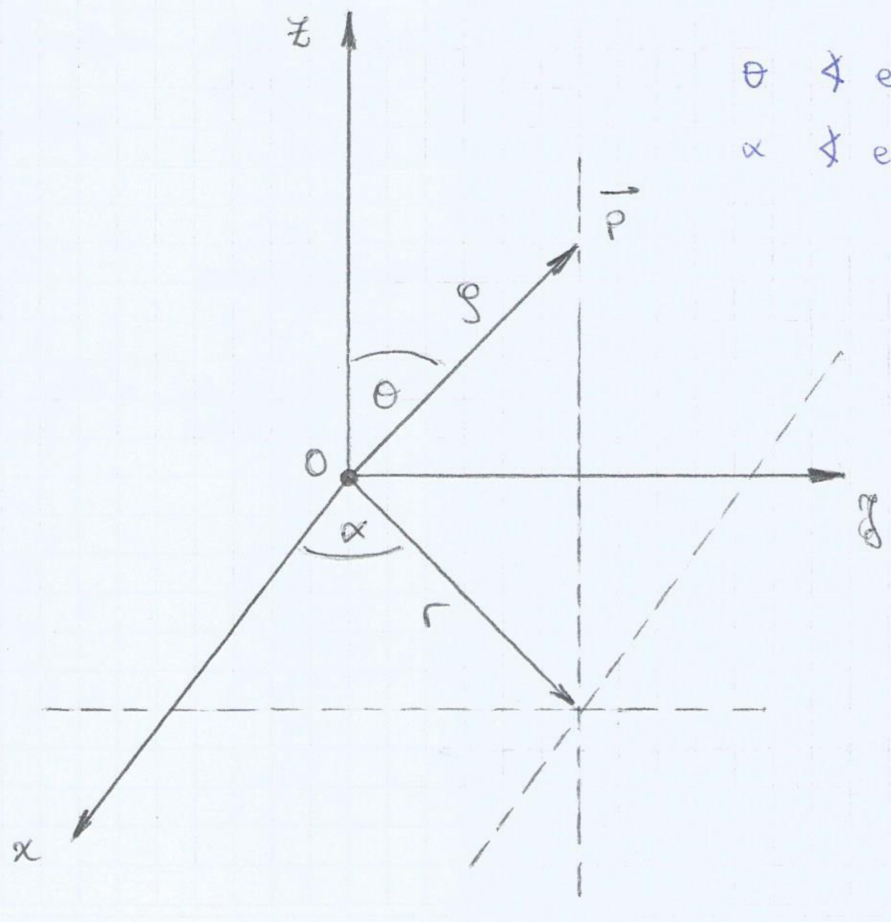
Coordenadas cilíndricas (α, r, z):



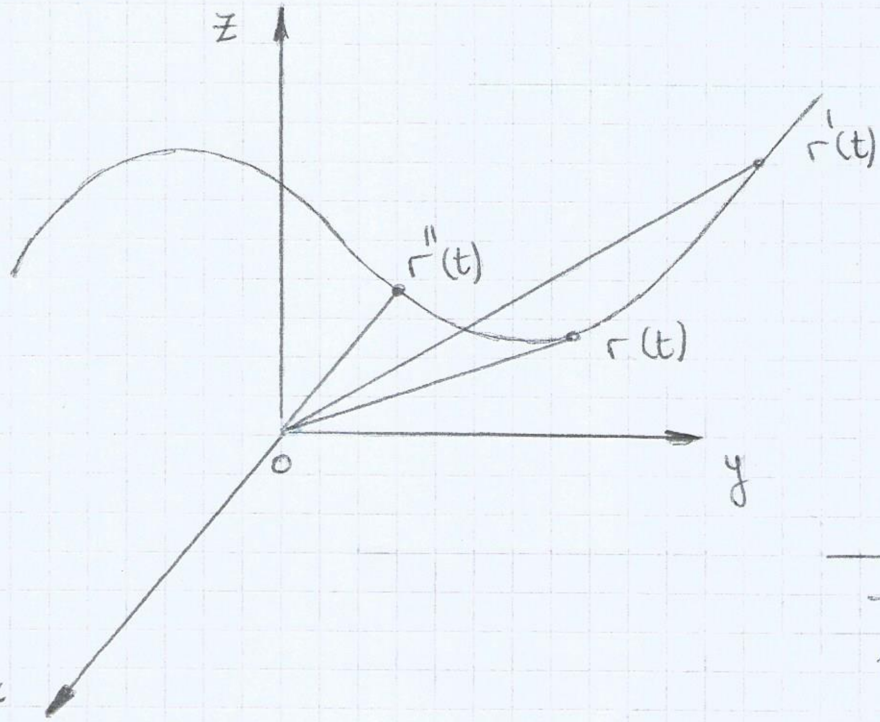
Coordenadas esféricas $(\rho, r, \theta, \alpha)$:

θ ∢ entre $\overline{\rho}$ y $\overline{z}$

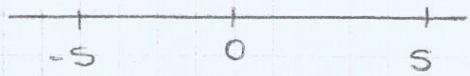
α ∢ entre $\overline{r}$ y $\overline{x}$



Cinemática:



trayectoria curvilínea.

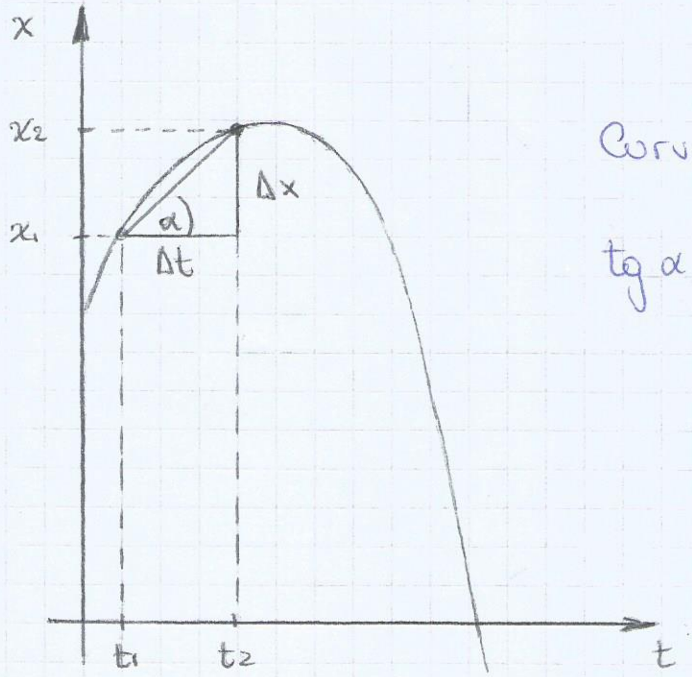


trayectoria rectilínea.

Velocidad: mide el cambio de posición en el tiempo.

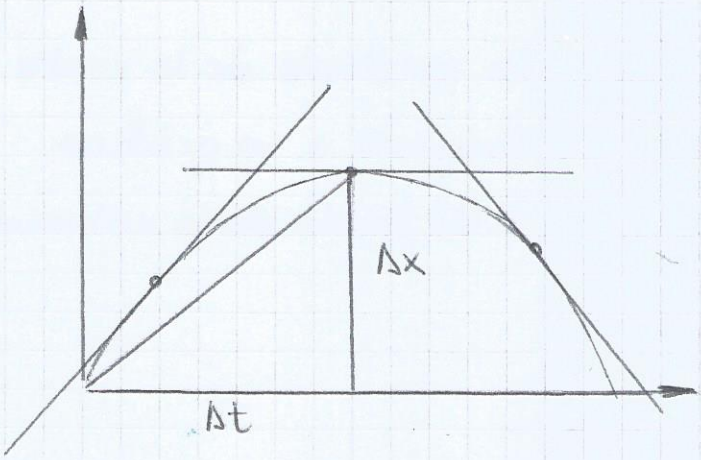
$$\bar{v}_p = \frac{\bar{s}(t_2) - \bar{s}(t_1)}{t_2 - t_1} = \frac{\Delta s}{\Delta t}$$ velocidad promedio

$$\bar{v}_I = \lim_{\Delta t \to 0} \frac{\bar{s}(t+\Delta t) - \bar{s}(t)}{\Delta t} = \frac{ds}{dt}$$ velocidad instantánea



Curva horaria

$$tg\,\alpha = \frac{x_2 - x_1}{t_2 - t_1} = \bar{v}_p$$



La velocidad en cada instante viene dada por la pendiente de la recta tg a la trayectoria.

Aceleración: mide el cambio de velocidad en el tiempo.

$$\bar{a}_p = \frac{v(t_2) - v(t_1)}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$ aceleración promedio

$$\bar{a}_I = \lim_{\Delta t \to 0} \frac{v(t+\Delta t) - v(t)}{\Delta t} = \frac{dv}{dt}$$ aceleración instantánea.

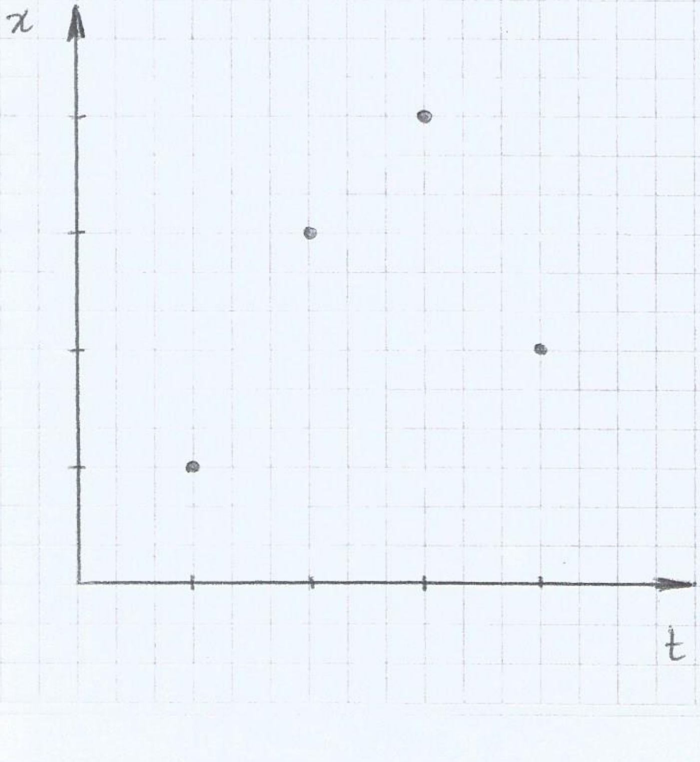
Movimiento rectilíneo:

$\bar{r}_1 = r(t_1) = x(t_1)\,\hat{x}$

$\bar{r}_2 = r(t_2) = x(t_2)\,\hat{x}$

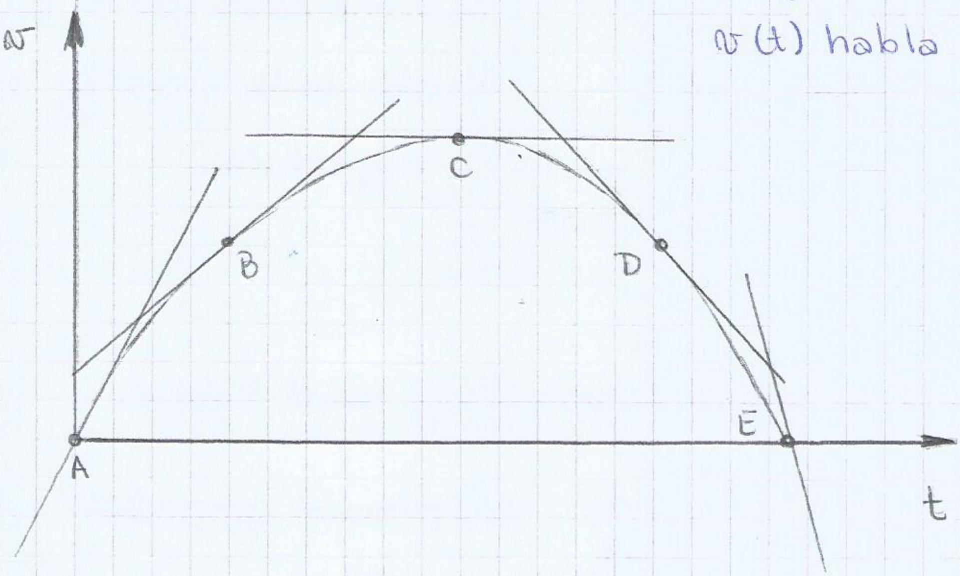
$\bar{r}_3 = r(t_3) = x(t_3)\,\hat{x}$

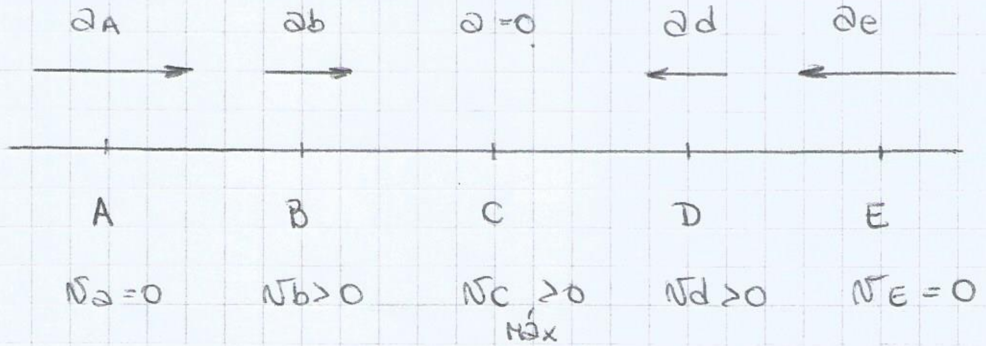
$\bar{r}_4 = r(t_4) = x(t_4)\,\hat{x}$



Supongamos que:

La pendiente de la recta tangente a un gráfico $v(t)$ habla de la velocidad





Si: $a>0$ y $v>0$ $\Rightarrow$ v aumenta en positivo

$a<0$ y $v<0$ $\Rightarrow$ v aumenta en negativo

$a>0$ y $v<0$ $\Rightarrow$

$a<0$ y $v>0$ $\Rightarrow$ v disminuye

$$v=\frac{dr}{dt}=\frac{dx}{dt}\hat{x}=v_x\hat{x}$$

$$a=\frac{dv}{dt}=\frac{d^2x}{dt^2}\hat{x}=a_x\hat{x}$$

NOTA

$v = \frac{dx}{dt}$ $\qquad \int_{x_0}^{x} dx = \int_{t_0}^{t} v\, dt \qquad$ con $x_0 = x(t_0)$

$$x(t) - x_0 = \int_{t_0}^{t} v\, dt$$

$$x(t) = x_0 + \int_{t_0}^{t} v\, dt$$

$a = \frac{dv}{dt}$ $\qquad \int_{v_0}^{v} dv = \int_{t_0}^{t} a\, dt \qquad$ con $v_0 = v(t_0)$

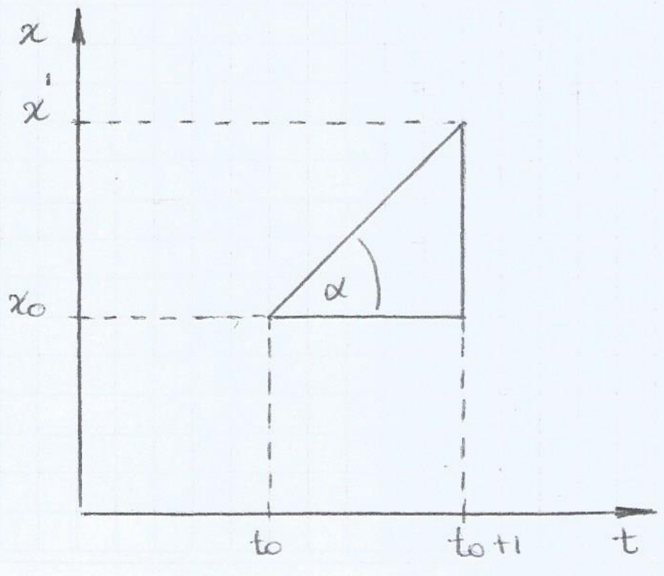
$$v(t) - v_0 = \int_{t_0}^{t} a\, dt$$

$$v(t) = v_0 + \int_{t_0}^{t} a\, dt$$

Movimiento Rectilineo Uniforme : $\bar{v} = cte$

$$x(t) = x_0 + \int_{t_0}^{t} v_c\, dt$$

$$x(t) = x_0 + v_c\,(t - t_0)$$



$x' = x\,(t_0 + 1)$

$$x(t_0+1) - x_0 = v_e\,(t_0+1-t_0)$$

$$v_e = \frac{x(t_0+1) - x_0}{t_0+1-t_0} = \operatorname{tg}\alpha$$

Movimiento Rectilineo Uniformemente variado $a_c = cte$

$$v(t) = v_0 + \int_{t_0}^{t} a_c\, dt$$

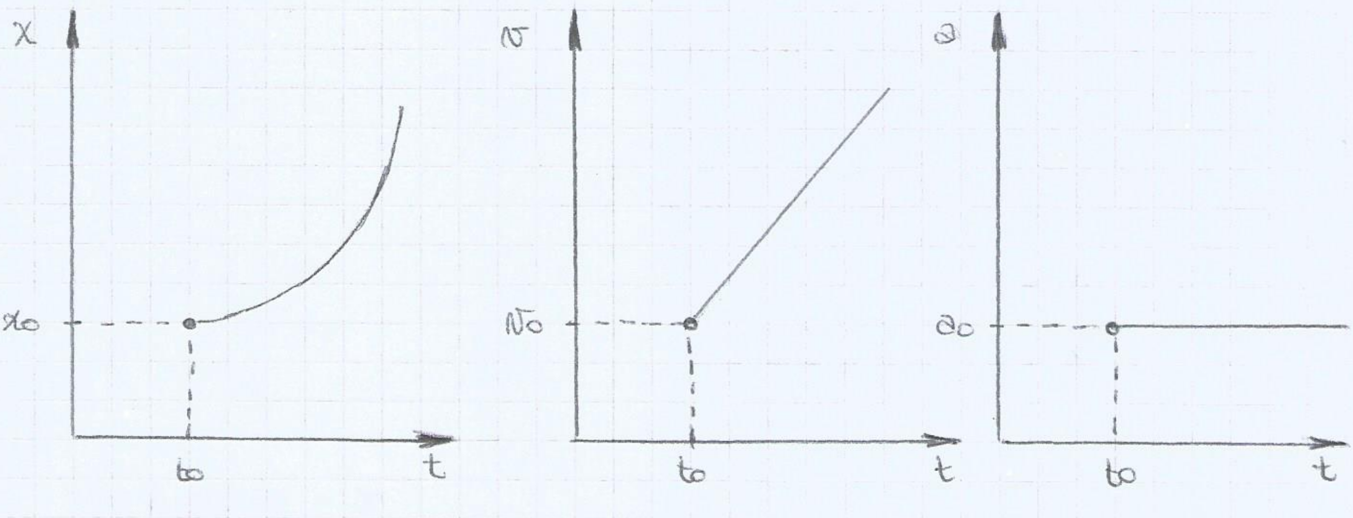
$$v(t) = v_0 + a_c\,(t-t_0)$$

$$x(t) = x_0 + \int_{t_0}^{t} v\, dt$$

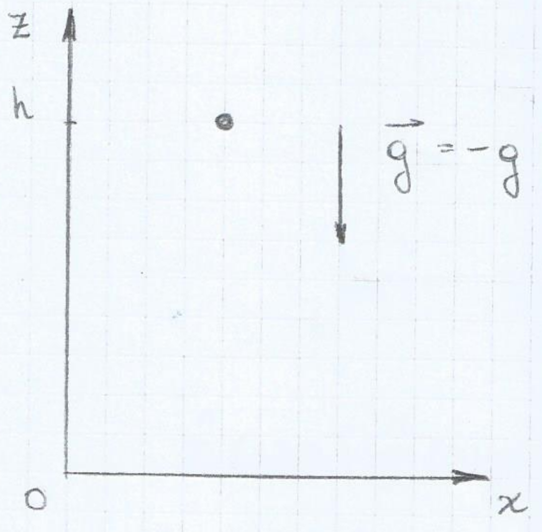
$$x(t) = x_0 + \int_{t_0}^{t} v_0 + a_c\,(t-t_0)\, dt$$

$$x(t) = x_0 + \int_{t_0}^{t} v_0\, dt + \int_{t_0}^{t} a_c\,(t-t_0)\, dt$$

$$x(t) = x_0 + v_0\,(t-t_0) + \frac{1}{2} a_c\,(t-t_0)^2$$



Caida libre:



$\vec{g} = -g\,\hat{z}$

Si : $t_0 = 0$

$v_0 = v(t_0) = 0$

$z_0 = h$

¿Cuánto tarda en caer?

$z(t) = h - \frac{1}{2} g\, t_c^2$ como vector $z\,\hat{z} : z_0\,\hat{z} + v_{0_z}(t - t_0)\hat{z} - \frac{1}{2} g (t - t_0)\hat{z}$

$0 = h - \frac{1}{2} g\, t_c^2$

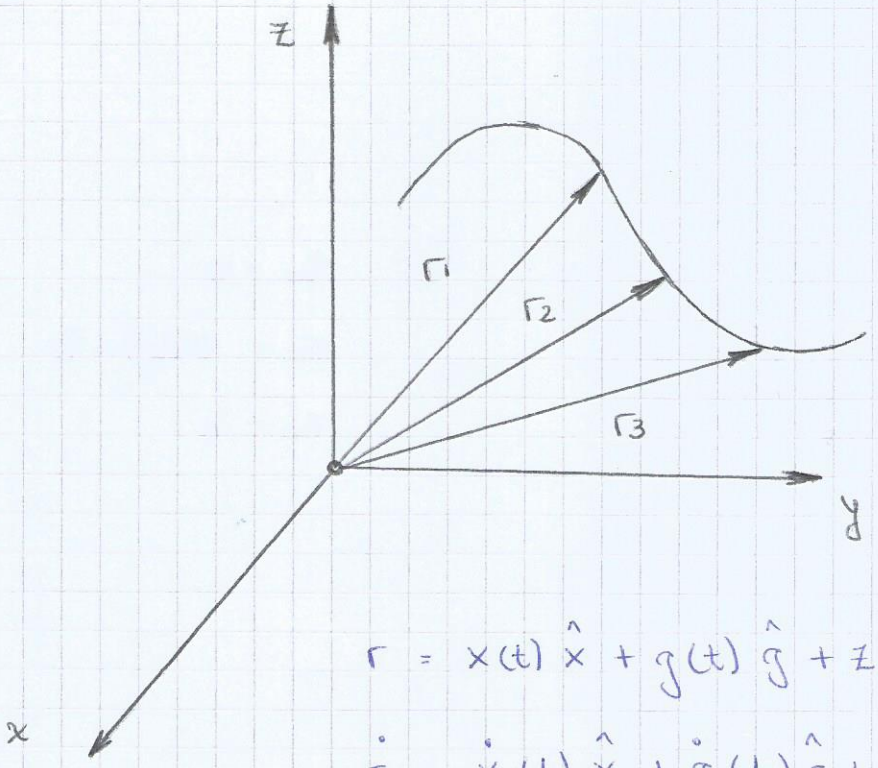
$t_c = \sqrt{\frac{2h}{g}}$

¿Con qué velocidad llega al piso?

$v(t)_z = -g\, t_c$

$v(t)_z = -g \left(\sqrt{\frac{2h}{g}}\right)$

$v(t)_z = -\sqrt{2gh}\;\hat{z}$

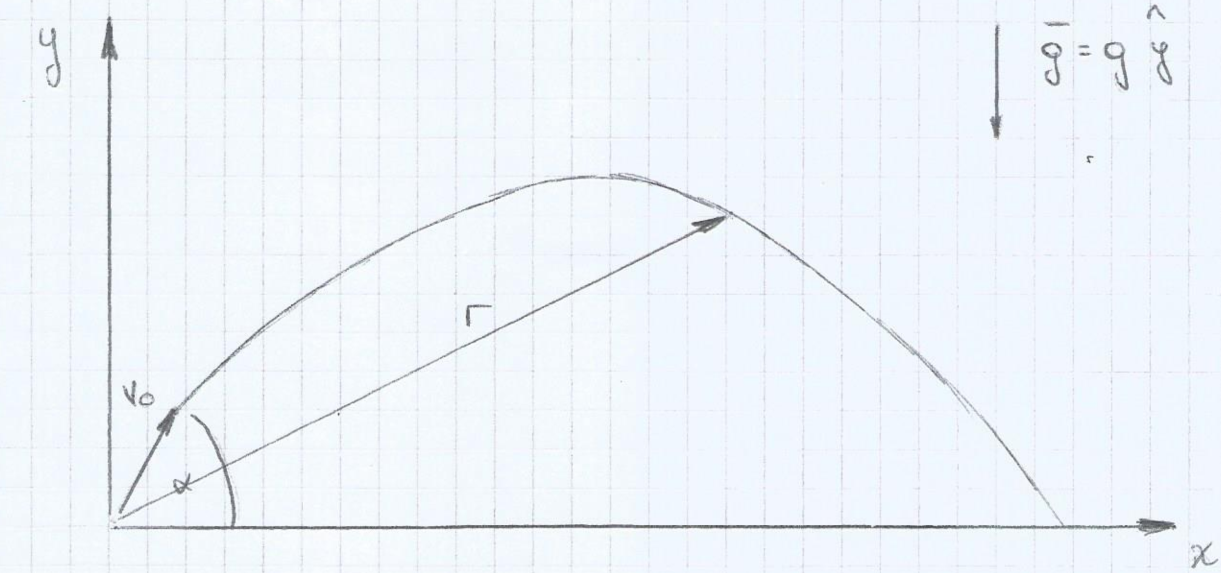


$$r = x(t)\,\hat{x} + y(t)\,\hat{y} + z(t)\,\hat{z}$$

$$\dot{r} = \dot{x}(t)\,\hat{x} + \dot{y}(t)\,\hat{y} + \dot{z}(t)\,\hat{z}$$

$$\ddot{r} = \ddot{x}(t)\,\hat{x} + \ddot{y}(t)\,\hat{y} + \ddot{z}(t)\,\hat{z}$$

Tiro oblicuo:



$$\bar{v}_0 = v_{0x}\,\hat{x} + v_{0y}\,\hat{y} \qquad v_0 = v_0 \cos\alpha\,\hat{x} + v_0 \operatorname{sen}\alpha\,\hat{y}$$

$$\bar{g} = -g\,\hat{y}$$

$$\bar{r} = x(t)\,\hat{x} + y(t)\,\hat{y}$$

Supongamos $t_0 = 0$, $x(t_0) = x_0 = 0$, $y(t_0) = y_0 = 0$

$$x(t) = x_0 + v_{0x}\,(t - t_0) \qquad y(t) = y_0 + v_{0y}(t-t_0) + \frac{1}{2} a_y (t-t_0)^2$$

$$x(t) = v_0 . \cos\alpha\; t \qquad y(t) = v_0 \,\mathrm{sen}\,\alpha\; t - \frac{1}{2} g\, t^2$$

$$a_y = -g\,\hat{y} \qquad v(t) = v_{0y} + a_y\,(t - t_0)$$

$$v(t) = v_0 \,\mathrm{sen}\,\alpha - g t$$

Ecuación de la trayectoria:

$$t = \frac{x(t)}{v_0 . \cos\alpha}$$

$$y(t) = v_0 \,\mathrm{sen}\,\alpha\; t - \frac{1}{2} g\, t^2$$

$$y(x) = v_0 \,\mathrm{sen}\,\alpha \left(\frac{x(t)}{v_0 . \cos\alpha}\right) - \frac{1}{2} g \left(\frac{x(t)}{v_0 \cos\alpha}\right)^2$$

$$y(x) = x(t)\, tg\,\alpha - \frac{g\, x(t)^2}{2 v_0^2} . \frac{1}{\cos^2\alpha}$$

Si $\dfrac{1}{\cos^2\alpha} = \dfrac{\mathrm{Sen}^2\alpha + \mathrm{Cos}^2\alpha}{\mathrm{Cos}^2\alpha} = tg^2\alpha + 1$

$$y(x) = x(t)\, tg\,\alpha - \frac{g\, x(t)^2}{2 v_0} (tg^2\alpha + 1)$$

$$y(x) = x(t)\, tg\,\alpha - \frac{g\, x(t)^2}{2 v_0}\, tg^2\alpha - \frac{g\, x(t)^2}{2 v_0}$$

$$0 = \left(y(x) - \frac{g\, x(t)^2}{2 v_0}\right) + x(t)\, tg\,\alpha - \frac{g\, x(t)^2}{2 v_0}\, tg^2\alpha$$

Ecuación de altura máxima: $(v_y = 0)$

$$v_y = v_0 \operatorname{sen} \alpha - gt$$

$$0 = v_0 . \operatorname{sen} \alpha - gt$$

$$t = \frac{v_0 . \operatorname{sen} \alpha}{g}$$ tiempo que tarda en alcanzar $y_{máx}$

$$y(t) = v_0 . \operatorname{sen} \alpha \, t - \frac{1}{2} g t^2$$

$$y_{máx} = v_0 . \operatorname{sen} \alpha \left(\frac{v_0 \operatorname{sen} \alpha}{g} \right) - \frac{1}{2} g \left(\frac{v_0 \operatorname{sen} \alpha}{g} \right)^2$$

$$y_{máx} = \frac{v_0^2 \operatorname{sen}^2 \alpha}{g} - \frac{1}{2} \cancel{g} \frac{v_0^2 \operatorname{sen}^2 \alpha}{g^{\cancel{2}}}$$

$$y_{máx} = \frac{1}{2} \frac{v_0^2 \operatorname{sen}^2 \alpha}{g}$$ (Altura máxima)

Ecuación de alcance: $(y = 0)$

$$y(t) = 0 = v_0 \operatorname{sen} \alpha \, t - \frac{1}{2} g t^2$$

$$0 = t \left(v_0 \operatorname{sen} \alpha - \frac{1}{2} g t \right)$$

$$t = 0 \qquad v_0 \operatorname{sen} \alpha - \frac{1}{2} g t = 0$$

$$v_0 \operatorname{sen} \alpha = \frac{1}{2} g t$$

$$t = \frac{2 v_0 \operatorname{sen} \alpha}{g}$$

$$x(t) = v_0 \cos\alpha \; t$$

$$x(t) = v_0 \cdot \cos\alpha \left(\frac{2 v_0 \operatorname{sen}\alpha}{g}\right)$$

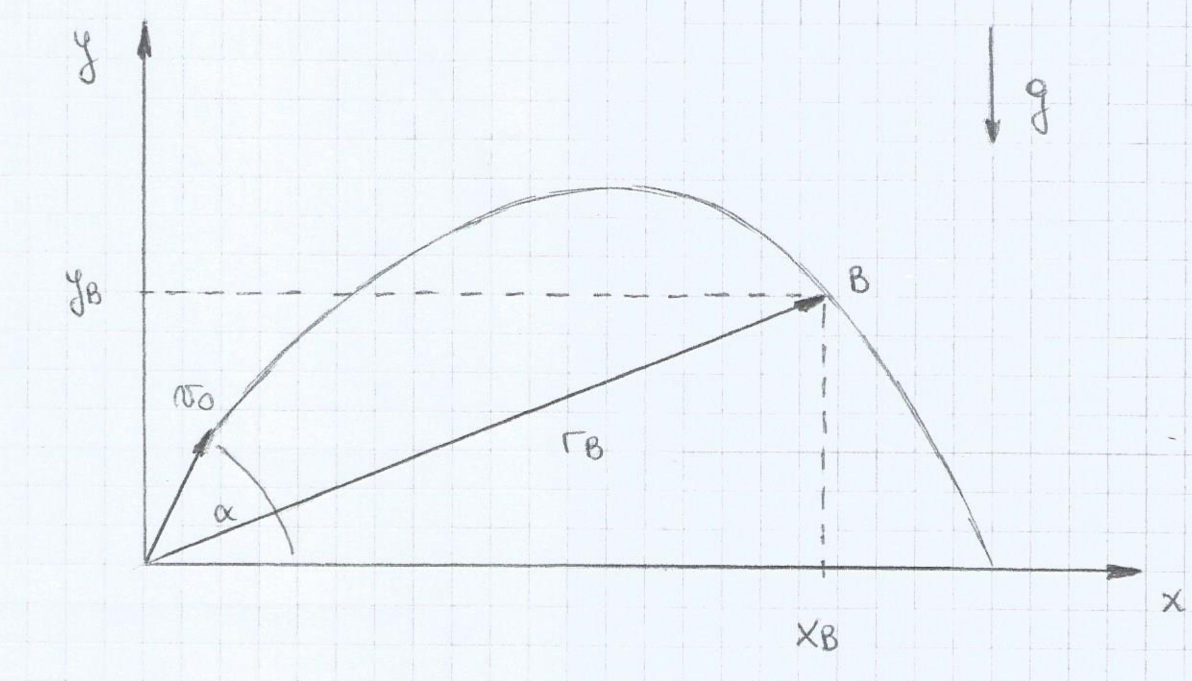
$$x(t) = \frac{2 v_0^2 \cos\alpha \operatorname{sen}\alpha}{g} \qquad \text{(Alcance)}$$

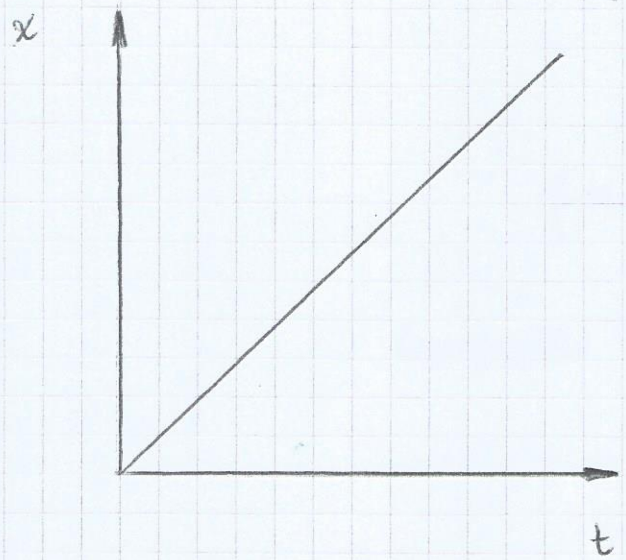
Ecuación del ángulo:

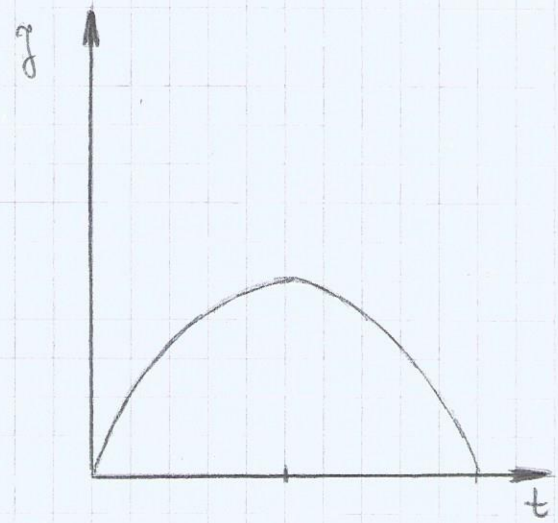
$$y_B = \operatorname{tg}\alpha \; x_B - \frac{g\, x_B^2}{2 v_0} (\operatorname{tg}^2\alpha + 1)$$

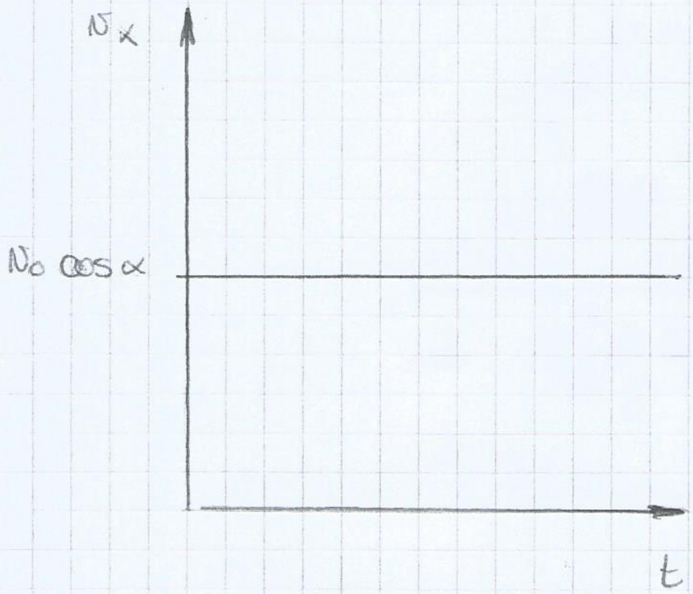
$$y_B = x_B \operatorname{tg}\alpha - \frac{g\, x_B^2}{2 v_0} \operatorname{tg}^2\alpha - \frac{g\, x_B^2}{2 v_0}$$

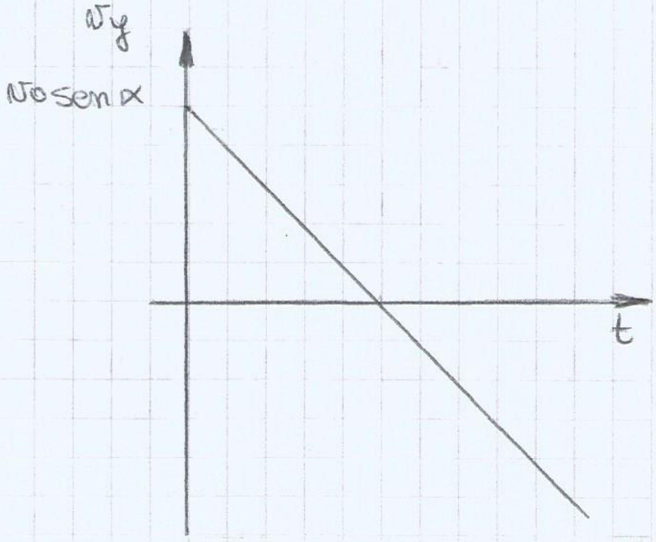
$$\operatorname{tg}\alpha = \frac{v_0^2}{g\, x_B} \pm \sqrt{\frac{v_0^4}{g^2 x_B^2} - \frac{2 v_0^2 y_B}{g\, x_B} - 1}$$

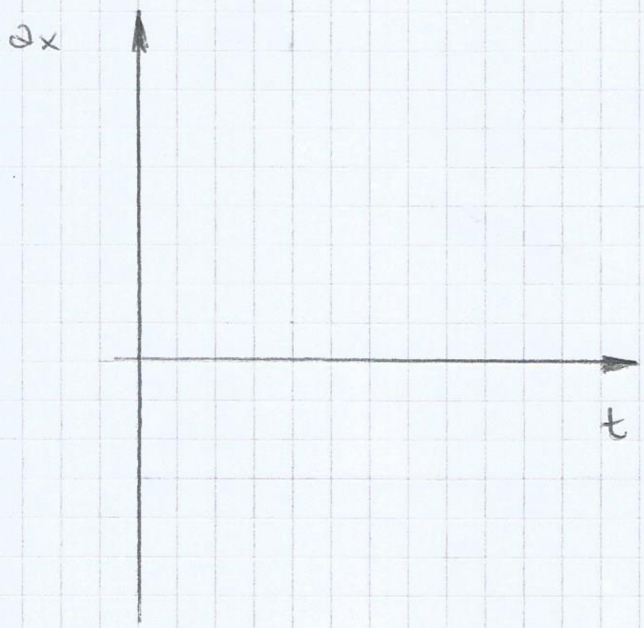


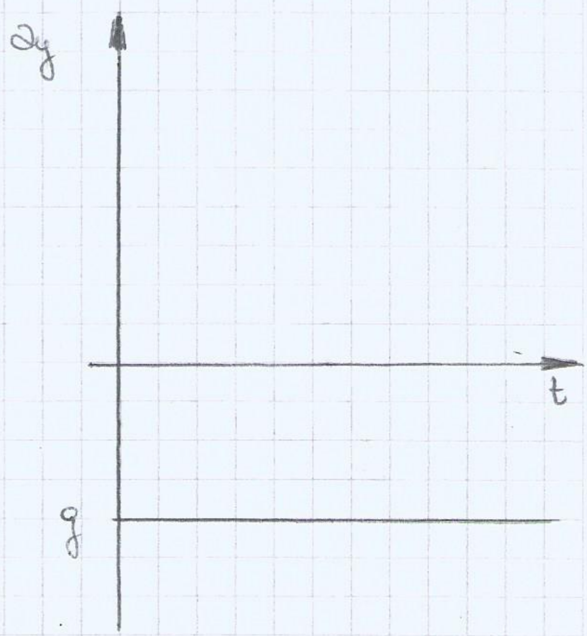






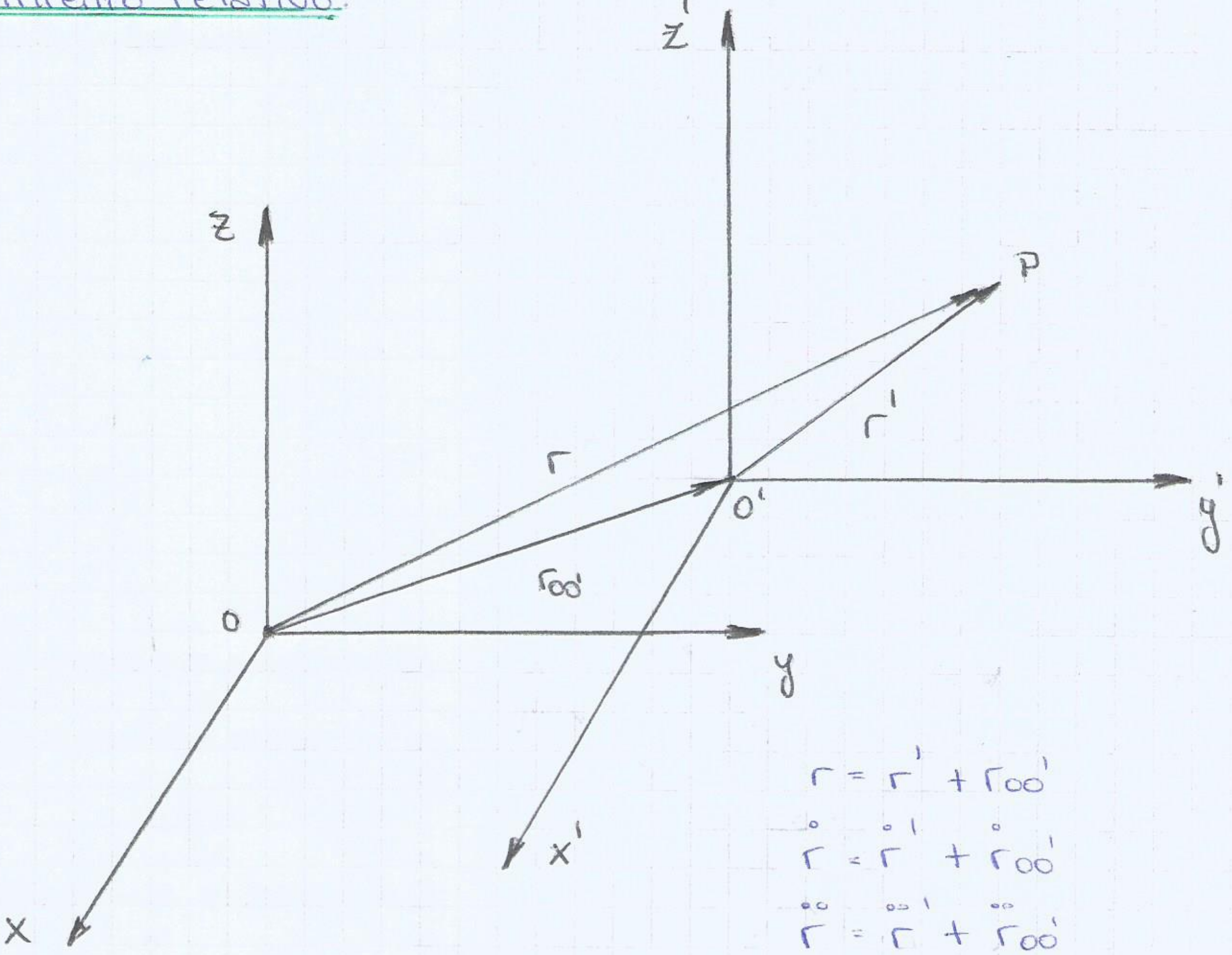






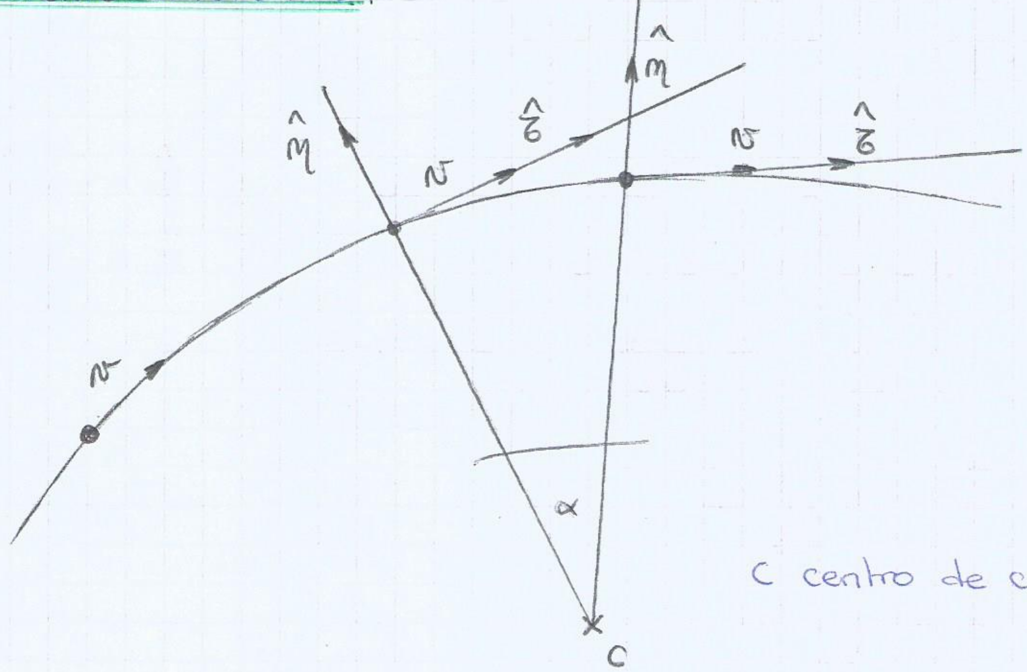
NOTA

Movimiento relativo:



$$r = r' + r_{oo'}$$
$$\dot{r} = \dot{r}' + \dot{r}_{oo'}$$
$$\ddot{r} = \ddot{r}' + \ddot{r}_{oo'}$$

Movimiento curvilineo:

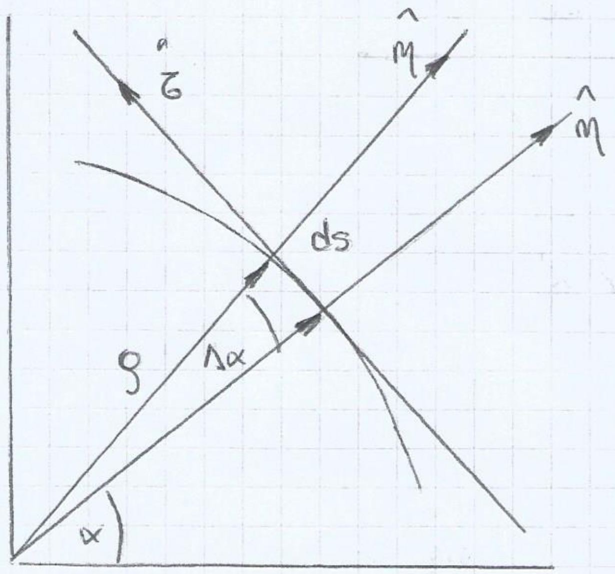


C centro de curvatura.

Hay una aceleración que cambia el módulo de v y otra componente de la aceleración que cambia la dirección v. Elijo un versor $\hat{\tau}$ que acompaña la dirección de la velocidad y otro versor $\hat{n}$.

$$\bar{v} = v\,\hat{\tau} \qquad\qquad \bar{a}_c \perp \bar{v}$$

$$\bar{a} = \underbrace{\dot{v}\,\hat{\tau}}_{\text{cambio de módulo}} + \underbrace{v\,\dot{\hat{\tau}}}_{\text{cambio de dirección y sentido.}}$$



$$\hat{\tau} = -\operatorname{sen}\alpha\,\hat{x} + \cos\alpha\,\hat{y}$$
$$\hat{n} = \cos\alpha\,\hat{x} + \operatorname{sen}\alpha\,\hat{y}$$

$$\dot{\hat{\tau}} = -\cos\alpha\,\dot{\alpha}\,\hat{x} - \operatorname{sen}\alpha\,\dot{\alpha}\,\hat{y}$$
$$\dot{\hat{\tau}} = -\dot{\alpha}\,(\cos\alpha\,\hat{x} - \operatorname{sen}\alpha\,\hat{y})$$
$$\dot{\hat{\tau}} = -\dot{\alpha}\,\hat{n}$$

$ds = \rho\,.\,d\alpha\,\hat{\tau}$ arco recorrido $\qquad \dot{\hat{\tau}} = -\dot{\alpha}\,\hat{n}$

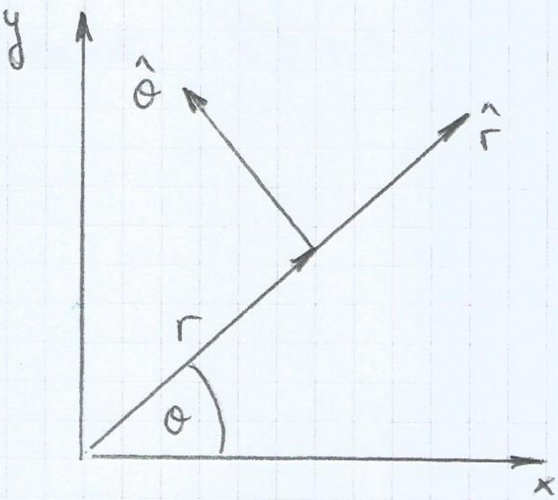
$$ds = v\,dt\,\hat{\tau} = \rho\,d\alpha\,\hat{\tau} \qquad\qquad \dot{\hat{\tau}} = -\frac{v}{\rho}\,\hat{n}$$

$$v = \rho\,\frac{d\alpha}{dt} \qquad v = \rho\,\dot{\alpha}$$

$$\bar{a} = \dot{v}\,\hat{\tau} - v.\frac{v}{\rho}\,\hat{n}$$

$$\bar{a} = \dot{v}\,\hat{\tau} - \frac{v^2}{\rho}\,\hat{n}$$

Aceleración en coordenadas polares:



$$\bar{r} = r\,\hat{r}$$

$$\hat{r} = \cos\theta\,\hat{x} + \operatorname{sen}\theta\,\hat{y}$$

$$\hat{\theta} = -\operatorname{sen}\theta\,\hat{x} + \cos\theta\,\hat{y}$$

$$\bar{v} = \dot{r}\,\hat{r} + r\,\dot{\hat{r}}$$

$$\bar{v} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta}$$

$$\dot{\hat{r}} = -\operatorname{sen}\theta\,\dot{\theta}\,\hat{x} + \cos\theta\,\dot{\theta}\,\hat{y}$$

$$\dot{\hat{r}} = \dot{\theta}\left(-\operatorname{sen}\theta\,\hat{x} + \cos\theta\,\hat{y}\right)$$

$$\dot{\hat{r}} = \dot{\theta}\,\hat{\theta}$$

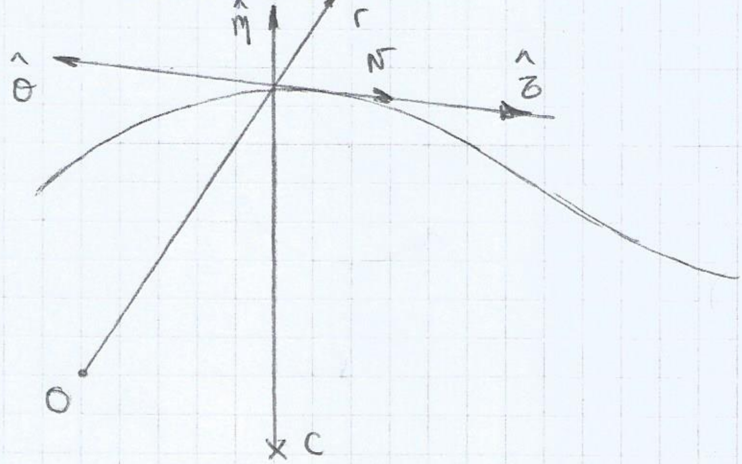
$$\bar{a} = \ddot{r}\,\hat{r} + \dot{r}\,\dot{\hat{r}} + \dot{r}\,\dot{\hat{r}} + r\,\ddot{\hat{r}}$$

$$\bar{a} = \ddot{r}\,\hat{r} + \dot{r}\,\dot{\hat{r}} + \dot{r}\dot{\theta}\,\hat{\theta} + r\ddot{\theta}\,\hat{\theta} + r\dot{\theta}\,\dot{\hat{\theta}}$$

$$\bar{a} = \ddot{r}\,\hat{r} + \dot{r}\dot{\theta}\,\hat{\theta} + \dot{r}\dot{\theta}\,\hat{\theta} + r\ddot{\theta}\,\hat{\theta} + r\dot{\theta} - \dot{\theta}\,\hat{r}$$

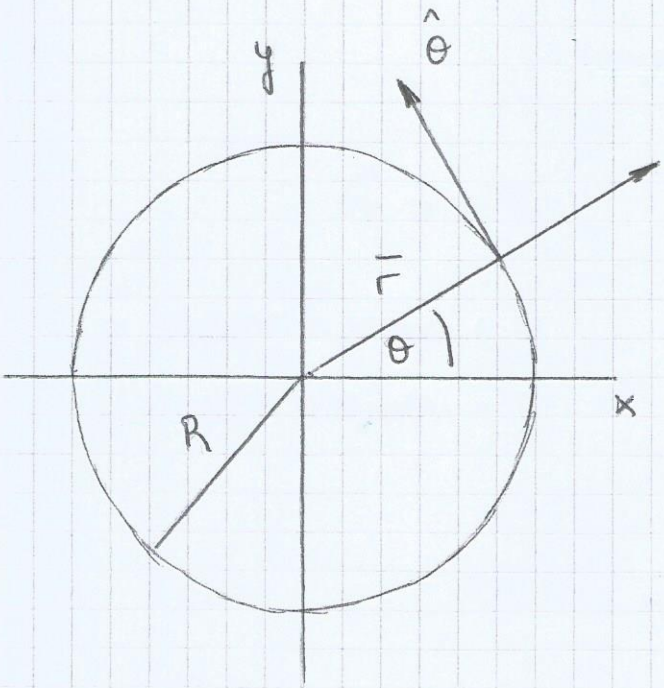
$$\bar{a} = \ddot{r}\,\hat{r} + 2\dot{r}\dot{\theta}\,\hat{\theta} + r\ddot{\theta}\,\hat{\theta} - \dot{\theta}^2 r\,\hat{r}$$

$$\bar{a} = \underbrace{\left(\ddot{r} - r\dot{\theta}^2\right)}_{\text{aceleración centrípeta}}\hat{r} + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\hat{\theta}$$



No coinciden las coordenadas polares con las coordenadas intrinsecas.

Movimiento Circular:



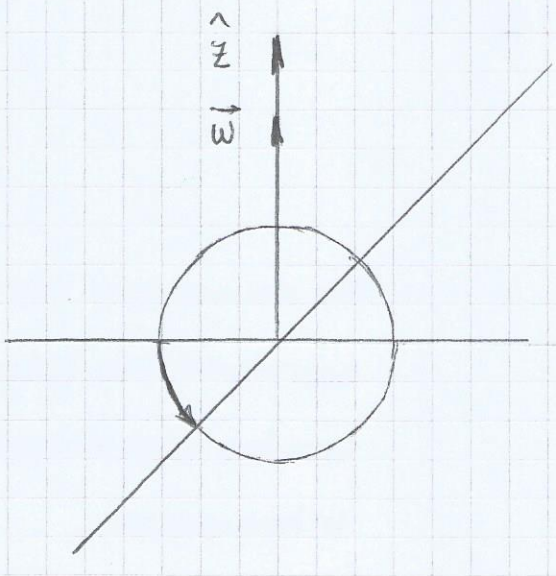
$\bar{r} = R\,\hat{r}$ $\quad R = cte$

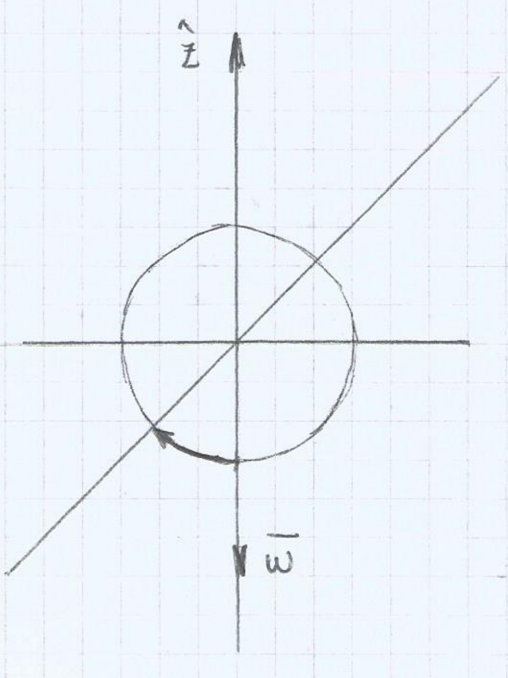
Velocidad angular: $\omega = \frac{d\theta}{dt}$ $\quad [\omega] = \frac{1}{[t]} = \frac{1}{seg}$

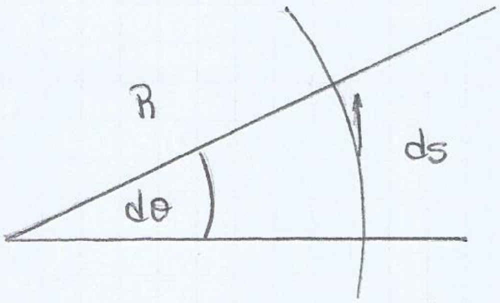
Aceleración angular: $\gamma = \frac{d^2\theta}{dt^2} = \frac{d\omega}{dt}$ $\quad [\gamma] = \frac{1}{[t]^2} = \frac{1}{seg^2}$

Velocidad angular (vector): como vector rotación $\vec{\omega}$

$|\vec{\omega}| = \omega$ dirección perpendicular al plano de rotación.







$ds = v\, dt$ $\qquad$ $ds = ds\,\hat{\theta}$

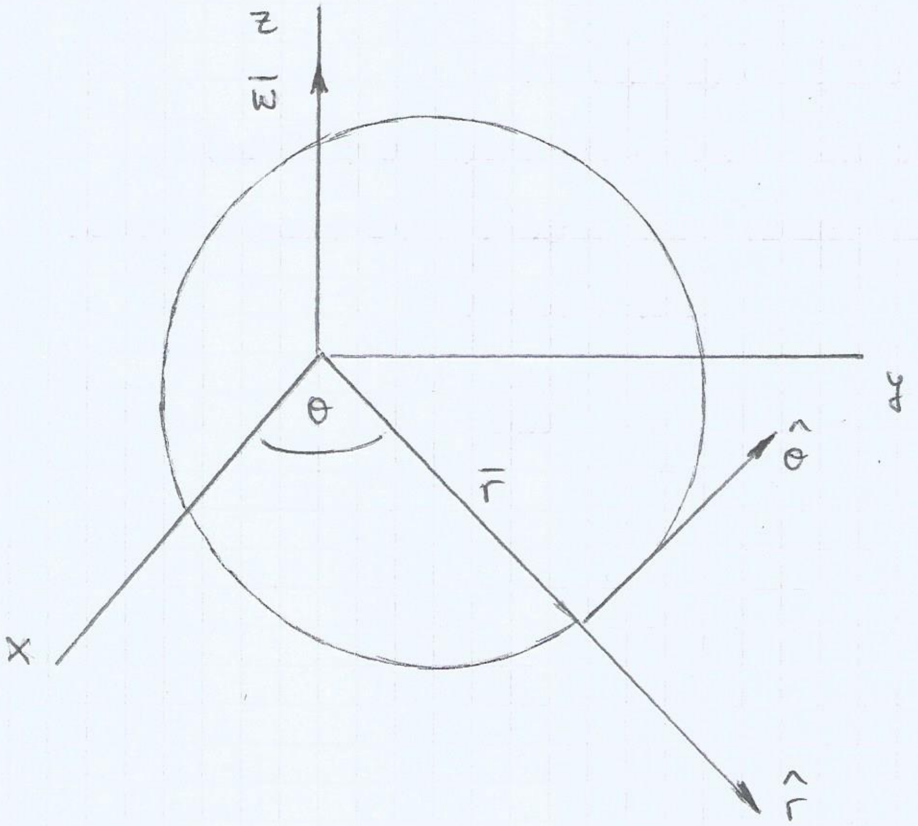
$ds = R \cdot d\theta$

$R\, d\theta\, \hat{\theta} = v \cdot dt$ $\qquad$ Como $\bar{v} = v \cdot \hat{\theta}$

$R\, d\theta\, \hat{\theta} = v\, dt\, \hat{\theta}$

$v = R \cdot \dfrac{d\theta}{dt}$

$v = R \cdot \omega$



$\bar{\omega} = \omega\, \hat{z}$

$\bar{v} = v \cdot \hat{\theta}$

$\bar{r} = R\, \hat{r}$

$\bar{v} = \bar{\omega} \times \bar{r}$

$\bar{v} = \omega R\, \hat{\theta}$

$\bar{a} = -R\dot{\theta}^2 \hat{r} + R\ddot{\theta}\,\hat{\theta}$

$\bar{a} = -R\omega^2 \hat{r} + R\gamma\,\hat{\theta}$

Periodo (T): es el tiempo que tarda en recorrer la circunferencia

Frecuencia (ν): la cantidad de vueltas por unidad de tiempo.

$$T = \frac{t}{n} = \frac{1}{\nu}$$

$$\theta(t) = \theta_0 + \omega_0 (t - t_0)$$

$$\theta(t+T) = \theta(t) + 2\pi$$

$$\theta(t+T) = \theta_0 + \omega_0 (t - t_0) + 2\pi$$

$$\theta(t+T) = \theta_0 + \omega_0 (t + T - t_0)$$

$$\theta(t+T) = \theta(t+T)$$

$$\theta_0 + \omega_0 (t - t_0) + 2\pi = \theta_0 + \omega_0 (t + T - t_0)$$

$$\cancel{\theta_0} + \cancel{\omega_0 t} - \omega_0 t_0 + 2\pi = \cancel{\theta_0} + \cancel{\omega_0 t} + \omega_0 T - \cancel{\omega_0 t_0}$$

$$2\pi = \omega_0 T$$

$$T = \frac{2\pi}{\omega_0}$$